

Encryption Technologies

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Cryptography

- “Cryptography” comes from the Greek words kryptos , meaning “concealed, hidden, veiled, secret, or mysterious,” and graphia , meaning “writing”; thus,
- cryptography is “the art of secret writing.”
- Cryptography is the practice of concealing information by converting plaintext (readable format) into ciphertext (unreadable format) using a key or encryption scheme
- It is the process of converting data into a scrambled code that is encrypted and sent across a private or public network
- Encryption is the method by which information is converted into secret code that hides the information's true meaning
- Encrypted messages can, at times, be decrypted by cryptanalysis (code breaking), even though modern encryption techniques are virtually unbreakable.

Objectives of Cryptography

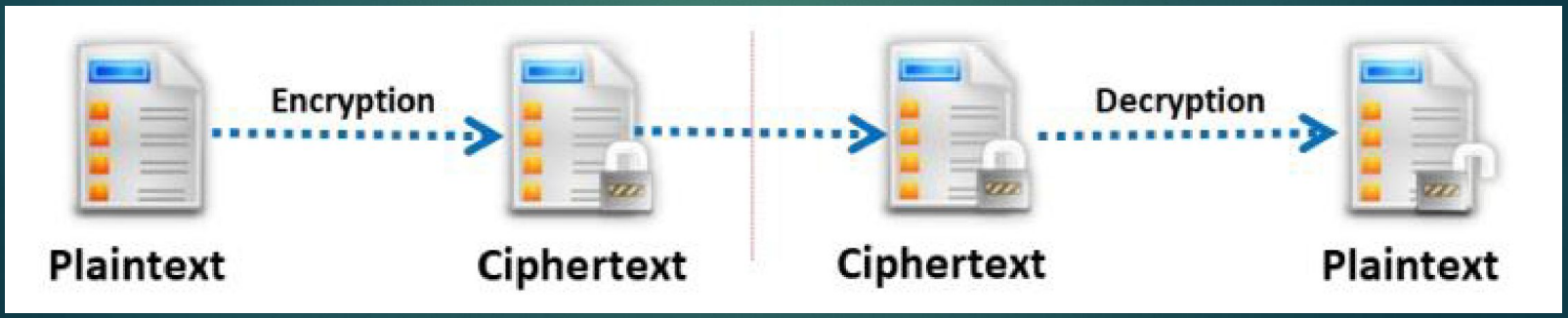
- Confidentiality : Assurance that the information is accessible only to those authorized to access it.
- Integrity : Trustworthiness of data or resources in terms of preventing improper and unauthorized changes.
- Authentication : Assurance that the communication, document, or data is genuine.
- Nonrepudiation : Guarantee that the sender of a message cannot later deny having sent the message and that the recipient cannot deny having received the message.

Cryptography Terminology

- Plain text Message in its original form (readable text)
- Cipher text Message in Coded Form
- Encryption Transforming Plain text to Cipher text
- Decryption transforming cipher text to plain text
- Encryption algorithm Method used for encryption

Cryptography Process

- The cryptography process involves transforming a Plaintext (readable format) into a cipher text using encryption algorithms
- On reaching the destination, is decrypt cipher text into readable plaintext

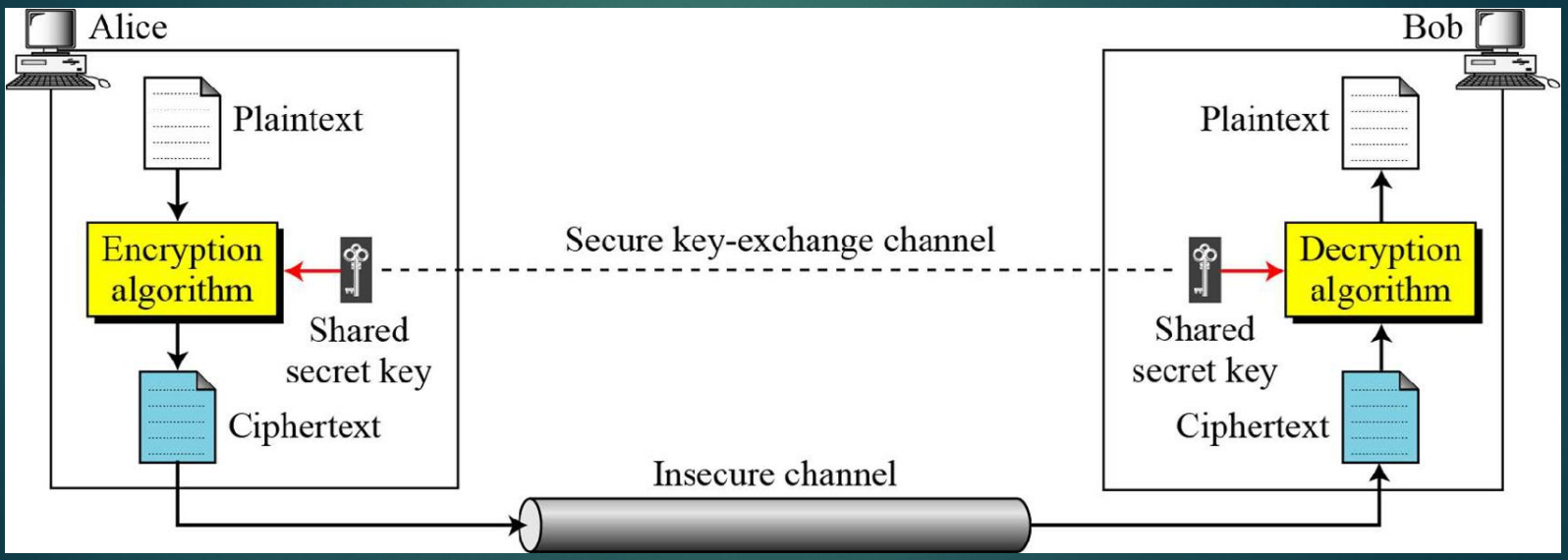


Types of Cryptography

- **Cryptography is categorized into two types according to the number of keys employed for encryption and decryption:**
  - **Symmetric encryption**
  - **Asymmetric Encryption**

Symmetric encryption

- **Symmetric encryption use the same pre-shared key (secret key) to encrypt and decrypt data.**
- **The sender uses a key to encrypt the plaintext and sends the resultant cipher text to the recipient, who uses the same key (used for encryption) to decrypt the cipher text into plaintext.**
- **Symmetric encryption is also known as secret-key cryptography, as it uses only one secret key to encrypt and decrypt the data.**


Alice
Bob
Plaintext
Plaintext
Encryption algorithm
Secure key-exchange channel
Decryption algorithm
Shared secret key
Shared secret key
Ciphertext
Ciphertext
Insecure channel

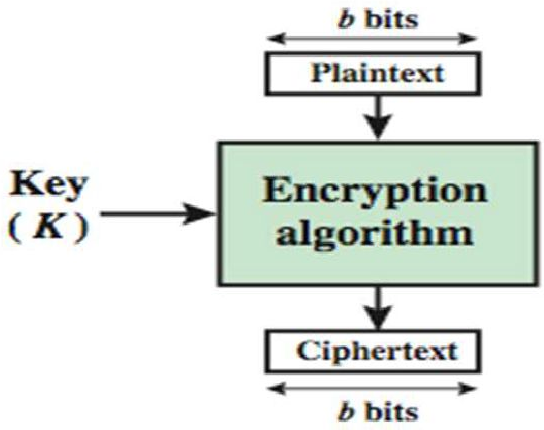
Symmetric encryption

- **Symmetric encryption comes in two forms, these are:**
- **Block cipher**
- **Stream ciphers**

Symmetric Encryption (Contd

Block Ciphers

- A block cipher is a method of encrypting data in blocks to produce cipher text using a cryptographic key and algorithm.
- The block cipher processes fixed size blocks simultaneously
- Block ciphers transform a fixed length block of plaintext into a common block of ciphertext of 64 or 128 bits.
- Common block ciphers include Data Encryption Standard(DES) with a 64 bit block size and Advanced Encryption Standard (AES) with a 128 bit block size.



(b) Block Cipher

Symmetric Encryption (Contd.)

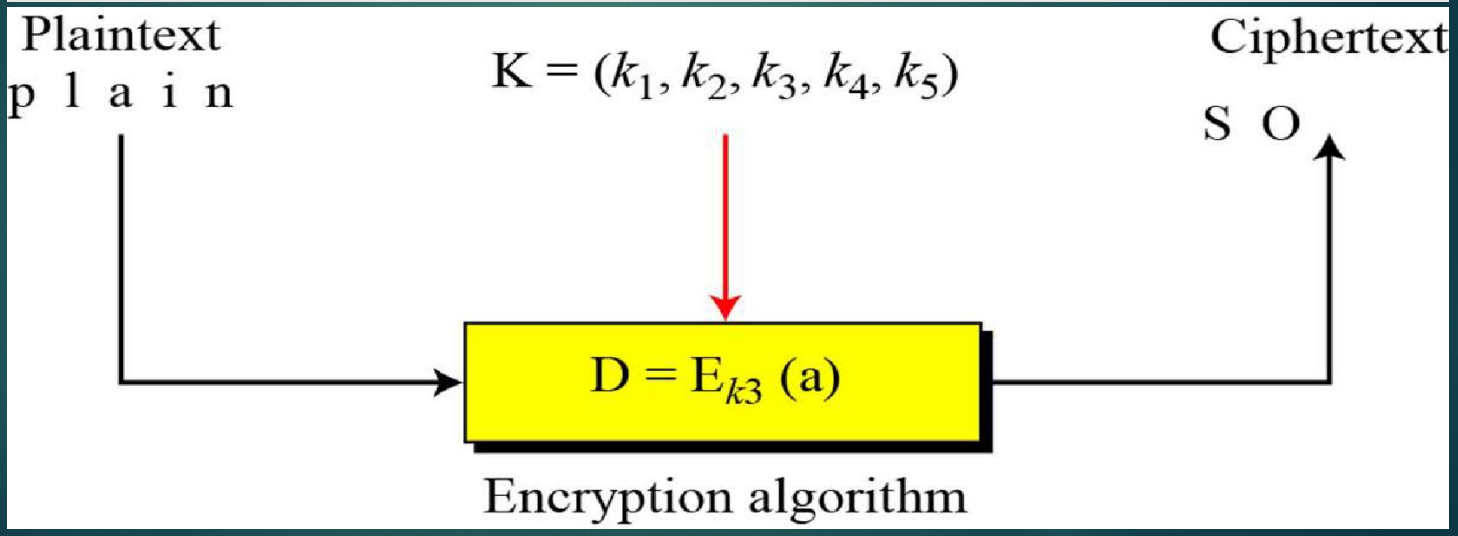
Stream Ciphers

- **Stream ciphersencrypt plaintext one byte or one bit at a time.**
- **Stream ciphers are basically a block cipher with a block size of one byte or bit.**
- **Stream ciphers are typically faster than block ciphers because data is continuously encrypted.**
- **Examples include RC4 and A5 which is used to encrypt Global System for Mobile communication(GSM)cell phone communications.**

Stream Cipher

$P = P_1P_2P_3, \ldots$ $C = C_1C_2C_3, \ldots$ $K = (k_1, k_2, k_3, \ldots)$

$C_1 = E_{k1}(P_1)$ $C_2 = E_{k2}(P_2)$ $C_3 = E_{k3}(P_3) \ldots$

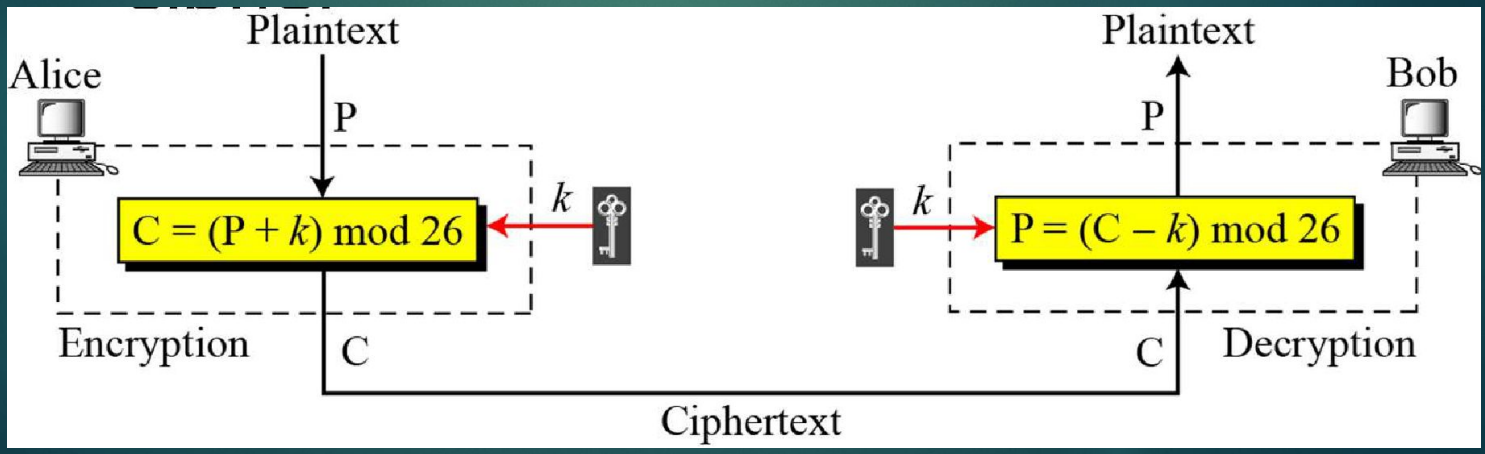


Stream Ciphers

- Additive ciphers can be categorized as stream ciphers in which the key stream is the repeated value of the key. In other words, the key stream is considered as a predetermined stream of keys or K = (k, k, …, k).
- The monoalphabetic substitution ciphers are also stream ciphers.
- However, each value of the key stream in this case is the mapping of the current plaintext character to the corresponding ciphertext character in the mapping table.

Additive Cipher

- The simplest monoalphabetic cipher is the additive cipher.
- This cipher is sometimes called a shift cipher and sometimes a Caesar cipher, but the term additive cipher better reveals its mathematical nature.
- When the cipher is additive, the plaintext, ciphertext, and key are integers in Z_{26}.



Example

- Use the additive cipher with key = 15 to encrypt the message “HELLO”.
- A-0, B-1,C-2……,Z-25

- Solution:
  We apply the encryption algorithm to the plaintext, character by character:

Monoalphabetic Substitution Cipher

- Because additive, multiplicative, and affine ciphers have small key domains, they are very vulnerable to brute-force attack.
- A better solution is to create a mapping between each plaintext character and the corresponding ciphertext character.
- Alice and Bob can agree on a table showing the mapping for each character.

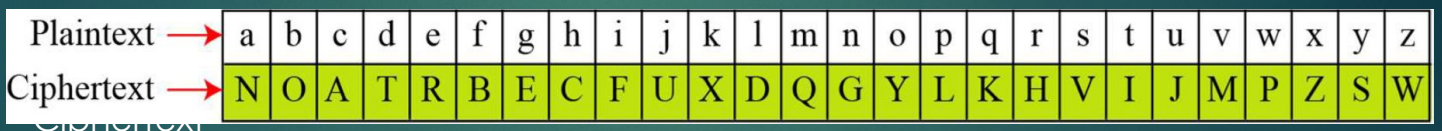
Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	N	O	A	T	R	B	E	C	F	U	X	D	Q	G	Y	L	K	H	V	I	J	M	P	Z	S	W

Example

- Encrypt the message

this message is easy to encrypt but hard to find the key

- Using this



Plaintext →	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext →	N	O	A	T	R	B	E	C	F	U	X	D	Q	G	Y	L	K	H	V	I	J	M	P	Z	S	W

- Ciphertext

ICFVQRVVNEFVRNVSIYRGAHSLIOJICNHTIYBFGTICRXRS

- Vigenere ciphers
- Vigenere ciphers are also stream ciphers according to the definition.
- In this case, the key stream is a repetition of m values, where m is the size of the keyword.
- In other words,

$$K = (k_1, k_2, \ldots k_m, k_1, k_2, \ldots k_m, \ldots)$$

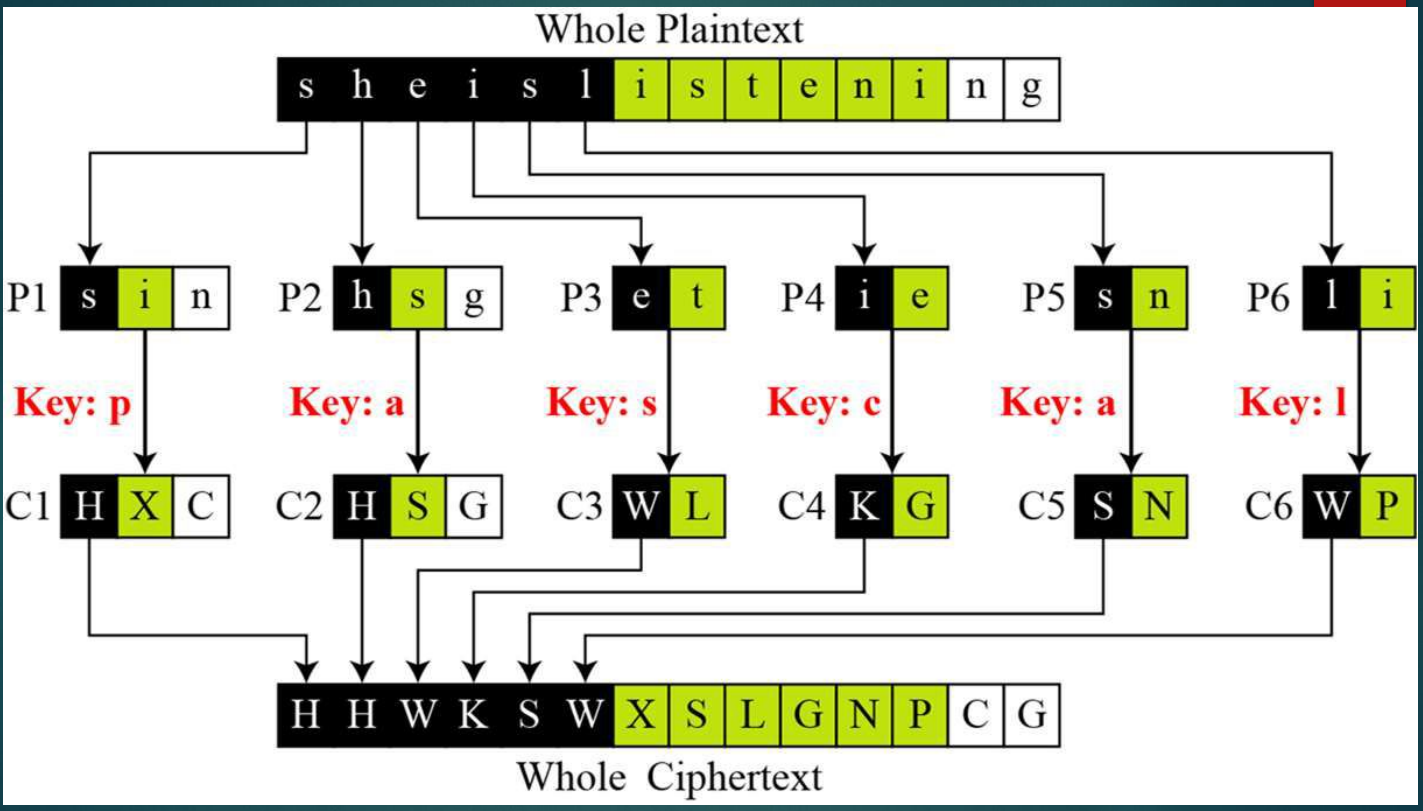
- In polyalphabetic substitution, each occurrence of a character may have a different substitute.
- The relationship between a character in the plaintext to a character in the ciphertext is one-to-many.

$$P = P_1P_2P_3 \ldots \qquad C = C_1C_2C_3 \ldots \qquad k = (k_1, P_1, P_2, \ldots)$$

$$\text{Encryption: } C_i = (P_i + k_i) \bmod 26 \qquad \text{Decryption: } P_i = (C_i - k_i) \bmod 26$$

Example

- **We can encrypt the message “She is listening” using the 6-character keyword “PASCAL”**


Whole Plaintext
s h e i s l i s t e n i n g
P1 s i n
P2 h s g
P3 e t
P4 i e
P5 s n
P6 l i
Key: p
Key: a
Key: s
Key: c
Key: a
Key: l
C1 H X C
C2 H S G
C3 W L
C4 K G
C5 S N
C6 W P
H H W K S W X S L G N P C G
Whole Ciphertext

	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	v	v	w	x	y	z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

Uses of symmetric encryption

- **Data at rest: Symmetric cryptography is also widely used to secure data stored in hard drives, archives, computers, flash drives, and laptops.**
- **Banking sector used Symmetric encryption to encrypt bulk data within the banking sector.**
- **Messaging apps like WhatsApp and signal uses end to end encryption to safeguard the privacy and confidentiality of communications on the platforms and also in authenticating users.**

- This type of cryptography works well when you are communicating with only a few people
- Because the sender and receiver must share the key before sending any messages, this technique is of limited use for the Internet, where individuals who have not had prior contact frequently require a secure means of communication.
- The solution to this problem is asymmetric encryption (public key cryptography).

Block Ciphers

- In a block cipher, a group of plaintext symbols of size m (m > 1) are encrypted together creating a group of ciphertext of the same size.
- A single key is used to encrypt the whole block even if the key is made of multiple values.

- Playfair ciphers are block ciphers. The size of the block is m = 2.
- Two characters are encrypted together.
- Hill ciphers are block ciphers. A block of plaintext, of size 2 or more is encrypted together using a single key (a matrix).
- Although the key is made of m × m values, it is considered as a single key.

Playfair Cipher

- not even the large number of keys in a monoalphabetic cipher provides security
- one approach to improving security was to encrypt multiple letters
- the Playfair Cipher is an example
- invented by Charles Wheatstone in 1854, but named after his friend Baron Playfair

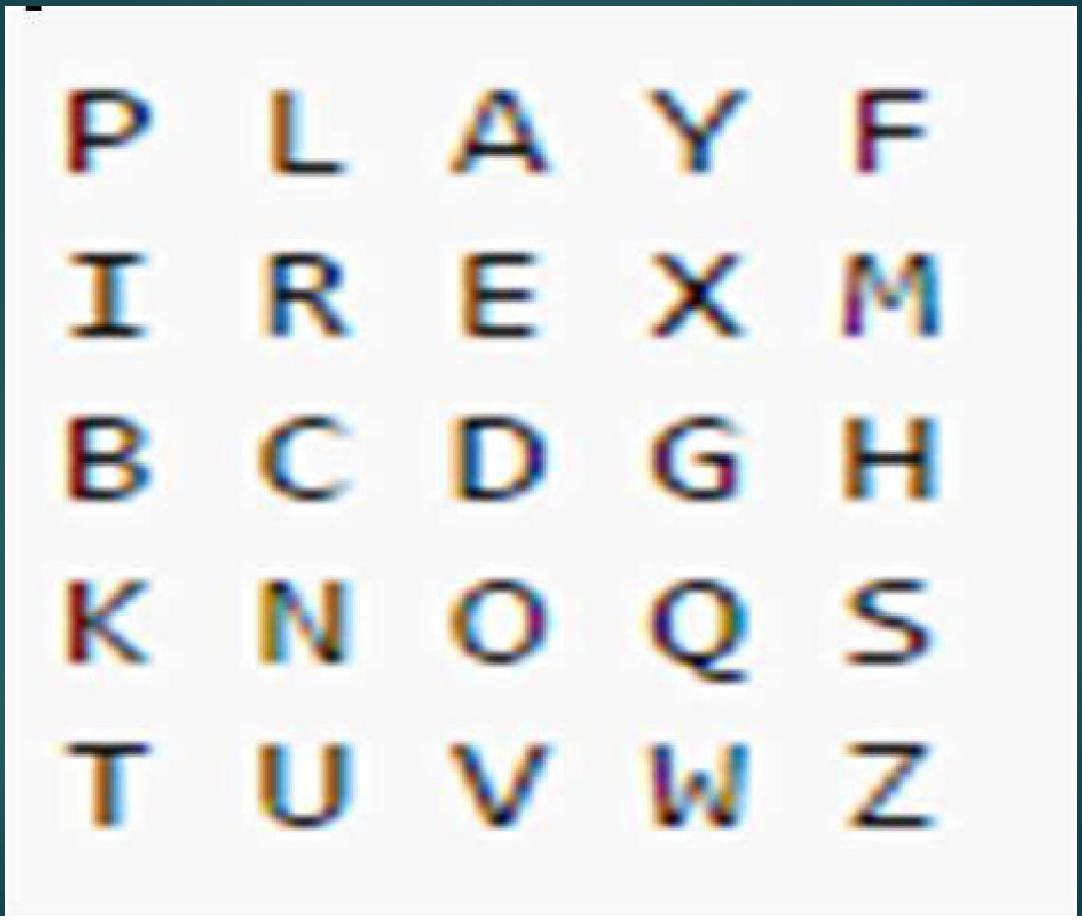
Playfair Key Matrix

- a 5X5 matrix of letters based on a keyword
- fill in letters of keyword (without duplicates)
- fill rest of matrix with other letters
- eg. using the keyword MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

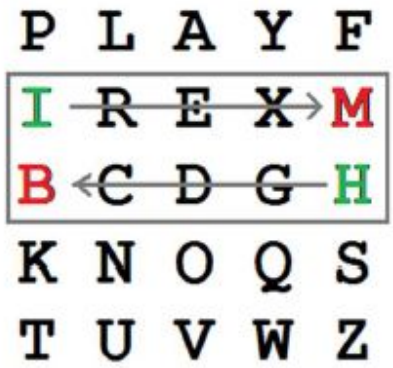
Example

- **Key: playfair**
- **Message "Hide the gold in the tree stump“**
- **(note the null "X" used to separate the repeated "E"s)**


P L A Y F
I R E X M
B C D G H
K N O Q S
T U V W Z

HI DE TH EG OL DI NT HE TR EX ES TU MP

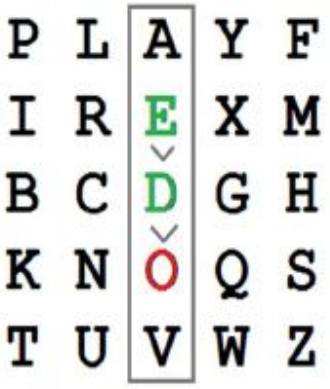
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HI

Shape: Rectangle
Rule: Pick Same Rows, Opposite Corners

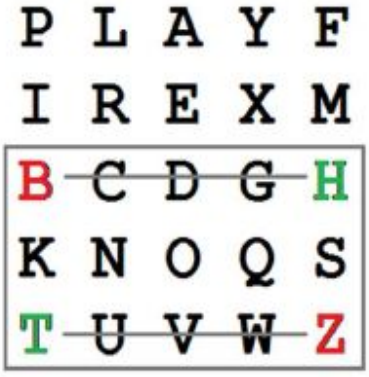
BM



DE

Shape: Column
Rule: Pick Items Below Each Letter, Wrap to Top if Needed

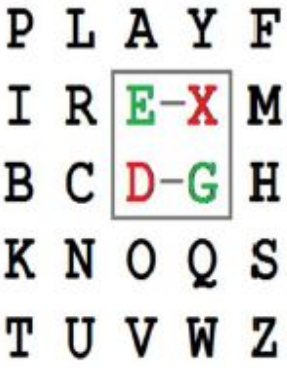
OD



TH

Shape: Rectangle
Rule: Pick Same Rows, Opposite Corners

ZB



EG

Shape: Rectangle
Rule: Pick Same Rows, Opposite Corners

XD

- 5. The pair OL forms a rectangle, replace it with NA
- 6. The pair DI forms a rectangle, replace it with BE
- 7. The pair NT forms a rectangle, replace it with KU
- 8. The pair HE forms a rectangle, replace it with DM
- 9. The pair TR forms a rectangle, replace it with UI
- 10. The pair EX forms a row, replace it with XM
- 11. The pair ES forms a rectangle, replace it with MO
- 12. The pair TU is in a row, replace it with UV
- 13. The pair MP forms a rectangle, replace it with IF

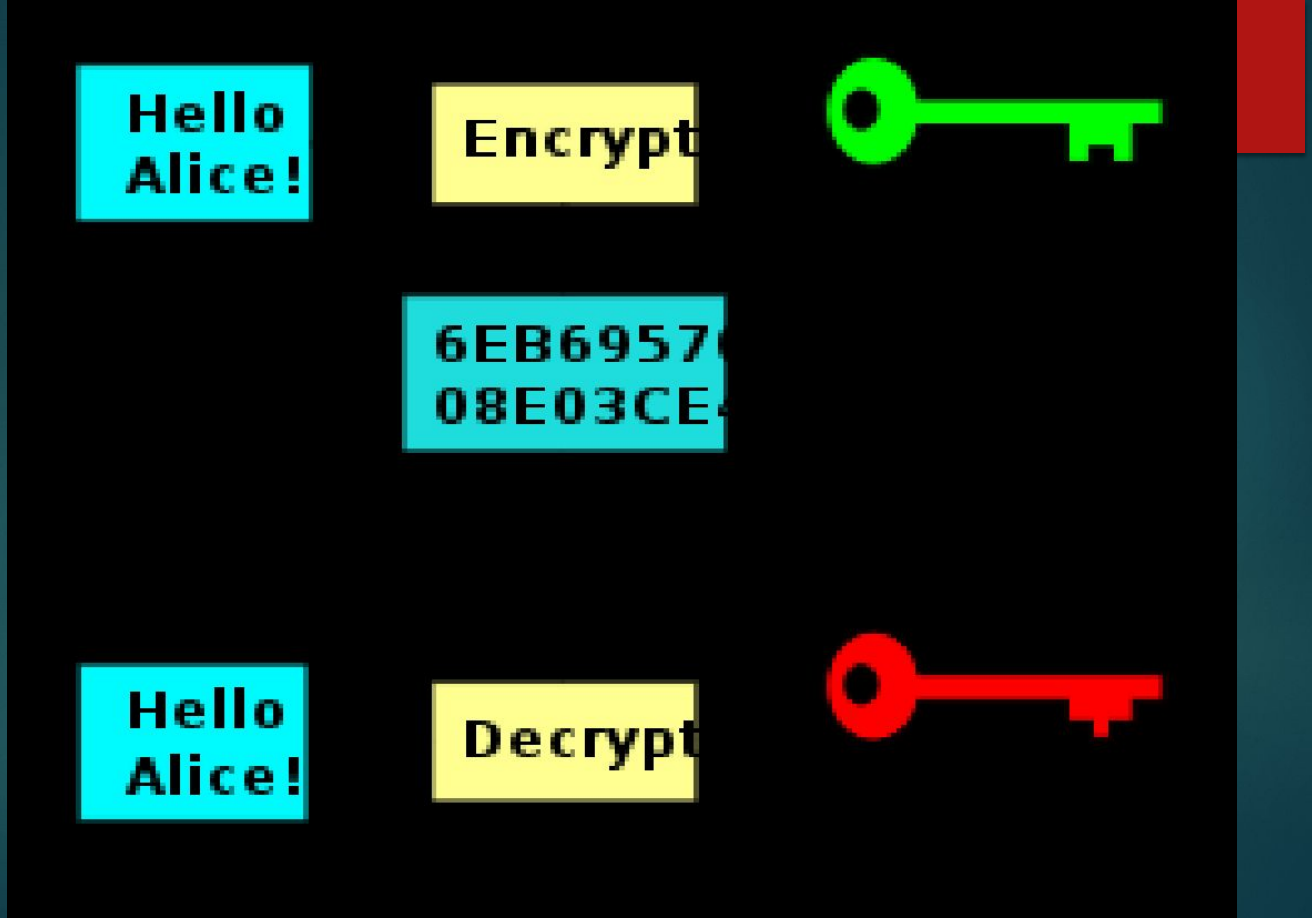
Asymmetric Encryption

- An asymmetric key system is an encryption method that uses a key pair comprising a public key available to anyone and a private key held only by the key owner, which helps to provide confidentiality, integrity, authentication, and nonrepudiation in data management.
- The concept of asymmetric encryption (also known as public key cryptography) was introduced to solve key management problems.
- Asymmetric encryption involves both a public key and a private key.

- Asymmetric encryption uses the following sequence to send a message :
  - An individual finds the public key of the person he or she wants to contact in a directory.
  - This public key is used to encrypt a message that is then sent to the intended recipient.
  - The receiver uses the private key to decrypt the message and reads it
- No one but the holder of the private key can decrypt a message encrypted with the corresponding public key.
- This increases the security of the information because all communications involve only public keys; the message sender never transmits or shares the private keys.
- The sender must link public keys with usernames in a secure manner to ensure that individuals claiming to be the intended recipient do not intercept the information.


Dear John,
This is my A/C
number
7974592830
Plain Text
Encryption
Public Key
Guuihifhofnk
bifkfnnfk
Nklclmlm
#^*&(*)_|_
Cipher Text
Decryption
Private key
Dear John,
This is my A/C
number
7974592830
Plain Text

- Common examples of asymmetric encryption algorithms
  - Digital Signature Standard (DSS) and Digital Signature Algorithm (DSA
  - Diffie Hellman (DH
  - Elliptic curve techniques


Hello Alice!
Encrypt
6EB6957
08E03CE
Hello Alice!
Decrypt

Why Public-Key Cryptography?

- developed to address two key issues:
  - key distribution – how to have secure communications in general without having to trust a KDC with your key.
  - digital signatures – how to verify a message comes intact from the claimed sender
- public invention due to Whitfield Diffie & Martin Hellman at Stanford University in 1976
- known earlier in classified community